Megan Bedell

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Major Research Interests: precision measurement techniques for detecting Earth-like planets; stellar astrophysics; statistical and computational methods for time-series data

Positions Held

2024 – present | **Research Scientist**, Center for Computational Astrophysics, Flatiron Institute

2019 – 2024 | **Associate Research Scientist**, Center for Computational Astrophysics, Flatiron Institute

2017 – 2019 | **Flatiron Research Fellow**, Center for Computational Astrophysics, Flatiron Institute

Education

2012 – 2017 | **University of Chicago**
Ph.D. in Astronomy & Astrophysics
Dissertation Title: Illuminating the Origins of Planets with Solar Twins
Advisor: Jacob L. Bean

2008 – 2012 | **Haverford College**
B.S. with Honors in Physics and with Highest Honors in Astronomy

Leadership & Service

- Terra Hunting Experiment, 2021–present
  - Scientific Operations Manager (designated), responsible for planning and oversight of survey operations for a 10-year radial velocity campaign targeting nearby Sun-like stars
  - Co-leader, Target Selection subgroup
  - Leader, Data Reduction Working Group, 2021–2024, coordinating cross-institutional planning for the HARPS3 spectroscopic data reduction pipeline
  - Member, Science Team
- Scientific Leadership, Center for Computational Astrophysics, 2019–present
  - One of the senior scientists anchoring the Exoplanets and Astronomical Data research groups; responsibilities include shaping group research directions, mentoring postdoctoral fellows, and fostering cross-group collaboration
  - Active contributor to CCA postdoctoral and pre-doctoral hiring, including annual participation in application review & candidate evaluation processes as well as formal committee membership (Flatiron Research Fellow Hiring Committee, 2022–2024; Pre-Doctoral Scholar Hiring Committee, 2022–2023 & 2025–2026)
  - Member of committees shaping CCA institutional strategy, including the Director Search Committee (2020) and the Hiring Process Committee (2020–2023)
  - Co-founder, Exoplanets weekly meeting, 2024–present
  - Co-leader, Astronomical Data Group weekly meeting, 2019–present
  - Member, Flatiron Institute Committee for Diversity, Equity, and Inclusion, 2020–2023
  - Organizer, CCA Colloquium, 2021–2024
- Member, Keck Planet Finder Science Team, 2020–present
- Science Organizing Committee member for **14** conferences/workshops to date, including:
  - *Sun-as-a-Star Workshop* (2023, NYC; SOC chair)
  - Three conferences in the *EPRV* series
  - Two conferences in the *HoRSE* series
  - *Telluric Line Hack Week* (2019, NYC)
- Time Allocation Committee or Panel member for **4** NASA cycles to date, including serving as Panel Chair

- Grant reviewer for NASA, NSF, and others
- Active referee for AAS, MNRAS, A&A, Nature journals

Mentorship & Teaching

- Extensive experience mentoring postdoctoral fellows, graduate students, and undergraduate students (see **Current & Former Mentees**)
- External examiner on thesis committees at Columbia, NYU, Yale, SUNY Stony Brook, University of Delaware
- Mentor for Cientifico Latino GSMI, 2024
- Instructor for the *LSST Data Science Fellowship Program* Session 10, 2019
- Guest lecturer for graduate classes at Columbia University, CUNY Graduate Center, and Universidade de São Paulo

Contributions to Open Science

- Creator of the *gaia-kepler.fun* crossmatch database, used in >50 peer-reviewed publications to date
- Co-creator (with Annelies Mortier) of the gr8stars database, a public-access repository of high-resolution spectra for a *Gaia*-selected sample of bright Northern-hemisphere FGK stars
- Developer of the `wobble` open-source code in python + TensorFlow
- Contributor to astronomical open-source software including `eleanor`, `lightkurve`

Honors & Awards

- Outstanding Paper Award, International Astrostatistics Association *(2020)*
- Josephine DeKarman Fellow *(2016–2017)*
- Lewis & Clark Field Scholar in Astrobiology *(2016–2017)*
- Illinois Space Grant Fellow *(2016–2017)*
- University of Chicago Harper Dissertation Fellow *(2016–2017)*
- NSF Graduate Research Fellow *(2013–2016)*
- Phi Beta Kappa *(2012)*

Outreach & Press

- Speaker at Brooklyn Public Library's Night at the Library event (8500 attendees), 2026
- Featured panelist, SF Presents: Imagining Other Worlds, October 2024
- Interviewed for NPR All Things Considered, Nature, PBS News Hour, Scientific American.
- Featured profile for the Flatiron Scientist Spotlight.
- **3** first-author papers covered by press including Science News article, ESO press release [picked up by Associated Press]
- Creator of TESS Roulette interactive data viewer (`https://tess.casino`); 43000 pageviews
- Volunteer for solar viewing events with SF *Path to Totality* eclipse program in NYC and Austin, 2024
- Local organizer for NYC Dept of Education STEM Career Day, 2018
- Astronomy on Tap presenter in NYC & Chicago, co-founder of Chicago AoT branch, 2016–2018
- Presenter, Chicago Life Long Learning, 2015–2017
- Presenter, Adler Planetarium Space Visualization Lab, 2014–2017

Current & Former Mentees

- **Independent postdoctoral fellows** managed through the Flatiron Research Fellowship (FRF) program, including:
  - Daniel Yahalomi — 2025, now a Torres Fellow at MIT
  - Michael Palumbo — 2024–present
  - Ryan Rubenzahl — 2024–present
  - Jiayin Dong — 2022–2025, now faculty at UIUC

  - Lionel Garcia — 2023–2025
  - Lily L. Zhao — 2021–2024, now a Sagan Fellow at UChicago
  - … & informal advising of numerous additional postdocs across the Exoplanets and Astronomical Data groups.
- **Graduate students** supervised (primarily) through the CCA Visiting Pre-Doctoral Scholar program:
  - Niamh O'Sullivan (Oxford) — 2026
  - Katlyn Hobbs (Queens University Belfast) — 2025–2026
  - Caprice Phillips (The Ohio State University) — 2024, resulting in 1 publication
  - Isabel Angelo (UCLA) — 2023–2025, resulting in 2 publications
  - Arjun Savel (UMD College Park) — 2022–2023, resulting in 2 publications
  - Chris Lam (University of Florida) — 2022–2023, resulting in 1 publication
  - Quang Tran (UT Austin) — 2021–2022, resulting in 1 publication
  - Arvind Gupta (Penn State) — 2021–2022, resulting in 1 publication
  - Lily L. Zhao (Yale University) — 2019–2020, resulting in 1 publication
- **Masters students** supervised through the CUNY Masters in Astrophysics program:
  - Lyniya Edwards (CUNY) — 2025–present
- **Undergraduates** supervised through summer programs including the Simons–NSBP Scholars Program, CUNY AstroCOM, and the Barnard Summer Research Institute:
  - Nusrat Jahan (CUNY Hunter College) — 2022
  - Lianys Feliciano (CUNY Hunter College) — 2022
  - Joann Roberts (Chicago State University) — 2020, resulting in senior thesis
  - Ilana Doran (Barnard College) — 2019–2021
  - Angus Beane (University of Pennsylvania) — 2018, resulting in 1 publication

Selected Talks

- MIT Astrophysics Colloquium, March 2026
- BU Astrophysics Seminar, March 2026
- McMaster University Physics & Astronomy Colloquium, March 2026
- CITA Seminar, March 2026
- University of Pennsylvania Astronomy Colloquium, February 2026
- Yale University Astronomy Colloquium, February 2026
- Five Colleges Astronomy Department Colloquium, March 2025
- Know Thy Star, Know Thy Planet 2 (invited talk), February 2025 **[slides / watch online]**
- SONG 2024 Science Meeting (invited talk), September 2024
- Two HoRSEs (invited panelist), July 2024
- NASA Goddard ASD Colloquium, October 2023
- San Francisco State University Physics Colloquium, April 2023
- University of Hawaii IfA Colloquium, April 2023
- EPRV5, March 2023 **[watch online]**
- University of Pennsylvania Astro Seminar, April 2021
- University College London Seminar, February 2020
- Princeton IAS Seminar, September 2019
- Sagan Summer Workshop, July 2019 **[watch online]**
- American Museum of Natural History Colloquium, May 2019
- Gordon Research Conference on Origins of Solar Systems, May 2019
- Carnegie EPL Colloquium, March 2019 **[watch online]**
- Cool Stars 20 (contributed plenary talk), August 2018 **[slides online]**
- Princeton Computational Astrophysics Workshop Series, May 2018 **[materials online]**

Publications *view on ADS*

***Starred** entries are undergrad or graduate student-led papers on which M. Bedell acted as a senior author.*

First- and Second-Author Papers:

1. **Savel, A. B.***, **Bedell, M.**, Kempton, E. M.-R., et al. 2025, Peering into the black box: forward-modeling the uncertainty budget of high-resolution spectroscopy of exoplanet atmospheres, AJ **169**, 135.
2. Zhao, L. L., **Bedell, M.**, Hogg, D. W., et al. 2024, A Compact, Coherent Representation of Stellar Surface Variation in the Spectral Domain, ApJ **977**, 140.
3. **Lam, C.***, **Bedell, M.**, Zhao, L., et al. 2024, gaspery: Optimized Scheduling of Radial Velocity Follow-Up Observations for Active Host Stars, AJ **168**, 200.
4. **Angelo, I.***, **Bedell, M.**, Petigura, E., et al. 2024, A Data-driven Spectral Model of Main Sequence Stars in Gaia DR3, ApJ **974**, 43.
5. **Savel, A. B.***, **Bedell, M.**, & Kempton, E. M.-R. 2024, cortecs: A Python package for compressing opacities, JOSS, **9**, 6104
6. **Gupta, A.F.*** & **Bedell, M.**, 2023, Fishing for Planets: A Comparative Analysis of EPRV Survey Performance in the Presence of Correlated Noise, AJ, **168**, 29.
7. **Tran, Q.H.***, **Bedell, M.**, Foreman-Mackey, D., et al. 2023, Joint Modeling of Radial Velocities and Photometry with a Gaussian Process Framework, ApJ, **950**, 162
8. Luger, R., **Bedell, M.**, Foreman-Mackey, D., Crossfield, I., *et al.*, 2021, Mapping Stellar Surfaces III: An Efficient, Scalable, and Open-Source Doppler Imaging Model, arXiv e-prints
9. Gan, T., **Bedell, M.**, Wang, S., Foreman-Mackey, D., *et al.*, 2021, HD 183579b: A Warm Sub-Neptune Transiting a Solar Twin Detected by TESS, MNRAS, **507**, 2220
10. Pope, B., **Bedell, M.**, Callingham, J., Vedantham, H., *et al.*, 2020, No Massive Companion to the Coherent Radio-Emitting M Dwarf GJ 1151, ApJ, **890**
11. **Bedell, M.**, Hogg, D., Foreman-Mackey, D., Montet, B., & Luger, R., 2019, WOBBLE: A Data-Driven Analysis Technique for Time-Series Stellar Spectra, AJ, **158**, 164
12. Luger, R., **Bedell, M.**, Vanderspek, R., & Burke, C., 2019, TESS Photometric Mapping of a Terrestrial Planet in the Habitable Zone: Detection of Clouds, Oceans, and Continents, arXiv e-prints
13. **Bedell, M.**, Bean, J., Meléndez, J., Spina, L., *et al.*, 2018, The Chemical Homogeneity of Sun-Like Stars in the Solar Neighborhood, ApJ, **865**, 68
14. **Bedell, M.**, Bean, J., Meléndez, J., Mills, S., *et al.*, 2017, Kepler-11 Is a Solar Twin: Revising the Masses and Radii of Benchmark Planets via Precise Stellar Characterization, ApJ, **839**, 94
15. Meléndez, J., **Bedell, M.**, Bean, J., Ramírez, I., *et al.*, 2017, The Solar Twin Planet Search. V. Close-In, Low-Mass Planet Candidates and Evidence of Planet Accretion in the Solar Twin HIP 68468, Astronomy and Astrophysics, **597**
16. **Bedell, M.**, Meléndez, J., Bean, J., Ramírez, I., *et al.*, 2015, The Solar Twin Planet Search. II. A Jupiter Twin Around a Solar Twin, Astronomy and Astrophysics, **581**
17. **Bedell, M.**, Meléndez, J., Bean, J., Ramírez, I., *et al.*, 2014, Stellar Chemical Abundances: In Pursuit of the Highest Achievable Precision, ApJ, **795**, 23
18. **Bedell, M.**, Villaume, A., Weiss, L., Sliski, D., *et al.*, 2011, Monitoring Hα Emission and Continuum of UXORs: RR Tauri, AJ, **142**, 164

Other Papers:

19. Rubenzahl, R. A., Hattori, S., Särkkä, S., et al. 2026, Scalable Gaussian Processes for Integrated and Overlapping Measurements Via Augmented State Space Models, AJ in press.
20. Yahalomi, D. A., Lu, T., Armitage, P. J., et al. 2026, The Astrometric Resoeccentric Degeneracy: Eccentric Single Planets Mimic 2:1 Resonant Planet Pairs in Astrometry, ApJL **999**, 1, L9.
21. Casey, A. R., Wheeler, A., **Bedell, M.**, et al. 2026, A Constrained Linear Model for Continuum Normalization of Stellar Spectra, ApJ **998**, 2, 192.
22. Angelo, I.*, Petigura, E., & **Bedell, M.** 2026, Seeking Spectroscopic Binaries with Data-driven Models, ApJ **997**, 2, 238.

23. Behmard, A., Brinkman, C. L., Hattori, S., et al. 2025, A Link between Rocky Planet Density and Host Star Chemistry, AJ **170**, 5, 282.
24. Rampalli, R., Johnson, J. W., Ness, M. K., et al. 2025, A Galactic Perspective on the (Unremarkable) Relative Refractory Depletion Observed in the Sun, ApJ in review.
25. Sun, Q., Ji, C., Wang, S. X., et al. 2025, Planets Around Solar Twins/Analogs (PASTA): II. Chemical abundances, systematic offsets, and clues as to planet formation, A&A **701**, A107.
26. O'Sullivan, N. K., Aigrain, S., Cretignier, M., et al. 2025, MNRAS. Measuring the Sun's radial velocity variability due to supergranulation over a magnetic cycle, MNRAS **541**, 4, 3942.
27. Freckelton, A. V., Mortier, A., **Bedell, M.,** et al. 2025, gr8stars – I. A homogeneous spectroscopic study of bright FGKM dwarfs and a public library of their high-resolution spectra, MNRAS **540**, 2, 1786.
28. Vieytes, M. C., Zhao, L. L., & **Bedell, M.** 2025, The influence of chromospheric activity on line formation, ApJ **981**, 4.
29. Lu, Y., Colman, I., Sayeed, M., et al. 2025, Evidence of Truly Young high-α Dwarf Stars, AJ **169**, 168.
30. Godoy-Rivera, D., Mathur, S., García, R. A., et al. 2025, Kepler meets Gaia DR3: homogeneous extinction-corrected color-magnitude diagram and binary classification, A&A **696**, A243.
31. Hon, M., Huber, D., Li, Y., et al. 2024, Asteroseismology of the Nearby K-Dwarf σ Draconis using the Keck Planet Finder and TESS, ApJ **975**, 147.
32. Eisner, N. L., Grunblatt, S. K., Barragán, O., et al. 2024, Planet Hunters TESS. V. A Planetary System Around a Binary Star, Including a Mini-Neptune in the Habitable Zone, AJ **167**, 241.
33. Rampalli, R., Ness, M. K., Edwards, G. H., et al. 2024, The Sun Remains Relatively Refractory Depleted: Elemental Abundances for 17,412 Gaia RVS Solar Analogs and 50 Planet Hosts , ApJ **965**, 176.
34. Sayeed, M., Ness, M. K., Montet, B. T., et al. 2023, Many Roads Lead to Lithium: Formation Pathways For Lithium-Rich Red Giants, ApJ **964**, 42.
35. Zhao, L. L., Dumusque, X., Ford, E. B., et al. 2023, The Extreme Stellar-signals Project. III. Combining Solar Data from HARPS, HARPS-N, EXPRES, and NEID, AJ **166**, 173.
36. Ferreira, T., Meléndez, J., Lorenzo-Oliveira, D., Bean, J.L., *et al.*, 2023, A Jupiter analogue and a cold Super-Neptune orbiting the solar-twin star HIP 104045, arXiv e-prints.
37. Bonfanti, A., Gandolfi, D., Egger, J., Fossati, L., *et al.*, 2023, TOI-1055 B: Neptunian Planet Characterised With HARPS, TESS, and CHEOPS, A&A **671**, L8.
38. **Behmard, A.***, Ness, M., Cunningham, E., & **Bedell, M.**, 2023, Elemental Abundances of Kepler Objects of Interest in APOGEE DR17, AJ **165**, 178.
39. Moran, A., Mingarelli, C., **Bedell, M.**, & Good, D., 2023, Improving Distances to Binary Millisecond Pulsars with Gaia, ApJ **954**, 89.
40. Angus, R., Price-Whelan, A., Zinn, J., **Bedell, M.**, *et al.*, 2022, The 3D Galactocentric Velocities of Kepler Stars: Marginalizing Over Missing Radial Velocities, AJ **164**, 25.
41. Zhao, L., Fischer, D., Ford, E., Wise, A., *et al.*, 2022, The EXPRES Stellar Signals Project II. State of the Field in Disentangling Photospheric Velocities, AJ **163**, 171.
42. Gan, T., Lin, Z., Wang, S., Mao, S., *et al.*, 2022, TOI-530b: A Giant Planet Transiting an M-Dwarf Detected by TESS, MNRAS **511**, 83.
43. Spina, L., Sharma, P., Meléndez, J., **Bedell, M.**, *et al.*, 2021, Chemical Evidence for Planetary Ingestion in a Quarter of Sun-Like Stars, Nature Astronomy **5**, 1163.
44. Hedges, C., Hughes, A., Zhou, G., David, T., *et al.*, 2021, TOI-2076 and TOI-1807: Two Young, Comoving Planetary Systems Within 50 Pc Identified by TESS That Are Ideal Candidates for Further Follow Up, AJ **162**, 54.
45. David, T., Contardo, G., Sandoval, A., Angus, R., *et al.*, 2021, Evolution of the Exoplanet Size Distribution: Forming Large Super-Earths Over Billions of Years, AJ **161**, 265.
46. Anderson, S., Dittmann, J., Ballard, S., & **Bedell, M.**, 2021, Higher Compact Multiple Occurrence Around Metal-Poor M-Dwarfs and Late-K-Dwarfs, AJ **161**, 203.
47. **Zhao, L.***, Hogg, D., **Bedell, M.**, & Fischer, D., 2021, Excalibur: A Nonparametric, Hierarchical Wavelength Calibration Method for a Precision Spectrograph, AJ textbf161, 80.
48. Spina, L., Nordlander, T., Casey, A., **Bedell, M.**, *et al.*, 2020, How Magnetic Activity Alters What We Learn From Stellar Spectra, ApJ **895**, 52.
49. Montet, B., Feinstein, A., Luger, R., **Bedell, M.**, *et al.*, 2020, The Young Planet DS Tuc Ab Has a Low Obliquity, AJ

159, 112.

50. Vedantham, H., Callingham, J., Shimwell, T., Tasse, C., *et al.*, 2020, Coherent Radio Emission From a Quiescent Red Dwarf Indicative of Star-Planet Interaction, Nature Astronomy **4**, 577.
51. Angus, R., Morton, T., Foreman-Mackey, D., van Saders, J., *et al.*, 2019, Toward Precise Stellar Ages: Combining Isochrone Fitting With Empirical Gyrochronology, AJ **158**, 173.
52. Feinstein, A., Montet, B., Foreman-Mackey, D., **Bedell, M.**, *et al.*, 2019, Eleanor: An Open-Source Tool for Extracting Light Curves From the TESS Full-Frame Images, PASP **131**, 94502.
53. Blancato, K., Ness, M., Johnston, K., Rybizki, J., & **Bedell, M.**, 2019, Variations in α-element Ratios Trace the Chemical Evolution of the Disk, ApJ **883**, 34.
54. Kreidberg, L., Luger, R., & **Bedell, M.**, 2019, No Evidence for Lunar Transit in New Analysis of Hubble Space Telescope Observations of the Kepler-1625 System, ApJ **877**, L15.
55. Lorenzo-Oliveira, D., Meléndez, J., Yana Galarza, J., Ponte, G., *et al.*, 2019, Constraining the Evolution of Stellar Rotation Using Solar Twins, MNRAS **485**, L68.
56. Carlos, M., Meléndez, J., Spina, L., dos Santos, L., *et al.*, 2019, The Li-Age Correlation: The Sun Is Unusually Li Deficient for Its Age, MNRAS **485**, 4052.
57. Botelho, R., Milone, A., Meléndez, J., **Bedell, M.**, *et al.*, 2019, Thorium in Solar Twins: Implications for Habitability in Rocky Planets, MNRAS **482**, 1690.
58. Lorenzo-Oliveira, D., Freitas, F., Meléndez, J., **Bedell, M.**, *et al.*, 2018, The Solar Twin Planet Search. The Age-Chromospheric Activity Relation, A&A **619**, A73.
59. **Beane, A.***, Ness, M., & **Bedell, M.**, 2018, Actions Are Weak Stellar Age Indicators in the Milky Way Disk, ApJ **867**, 31.
60. Crossfield, I., Guerrero, N., David, T., Quinn, S., *et al.*, 2018, A TESS Dress Rehearsal: Planetary Candidates and Variables From K2 Campaign 17, The Astrophysical Journal Supplement Series **239**, 5.
61. Spina, L., Meléndez, J., Karakas, A., dos Santos, L., *et al.*, 2018, The Temporal Evolution of Neutron-Capture Elements in the Galactic Discs, MNRAS **474**, 2580.
62. dos Santos, L., Meléndez, J., **Bedell, M.**, Bean, J., *et al.*, 2017, Spectroscopic Binaries in the Solar Twin Planet Search Program: From Substellar-Mass to M Dwarf Companions, MNRAS **472**, 3425.
63. Gandolfi, D., Barragán, O., Hatzes, A., Fridlund, M., *et al.*, 2017, The Transiting Multi-Planet System HD 3167: A 5.7 $M_\oplus$ Super-Earth and an 8.3 $M_\oplus$ Mini-Neptune, AJ **154**, 123.
64. Malik, M., Grosheintz, L., Mendonça, J., Grimm, S., *et al.*, 2017, HELIOS: An Open-Source, GPU-accelerated Radiative Transfer Code for Self-Consistent Exoplanetary Atmospheres, AJ **153**, 56.
65. Barragán, O., Grziwa, S., Gandolfi, D., Fridlund, M., *et al.*, 2016, EPIC 211391664b: A 32 $M_\oplus$ Neptune-Size Planet in a 10 Day Orbit Transiting an F8 Star, AJ **152**, 193.
66. dos Santos, L., Meléndez, J., do Nascimento, J., **Bedell, M.**, *et al.*, 2016, The Solar Twin Planet Search. IV. The Sun as a Typical Rotator and Evidence for a New Rotational Braking Law for Sun-Like Stars, Astronomy and Astrophysics, **592**
67. Tucci Maia, M., Ramírez, I., Meléndez, J., **Bedell, M.**, *et al.*, 2016, The Solar Twin Planet Search. III. The [Y/Mg] Clock: Estimating Stellar Ages of Solar-Type Stars, A&A **590**, A32.
68. Ramírez, I., Meléndez, J., Bean, J., Asplund, M., *et al.*, 2014, The Solar Twin Planet Search. I. Fundamental Parameters of the Stellar Sample, A&A **572**, A48.
69. Meléndez, J., Ramírez, I., Karakas, A., Yong, D., *et al.*, 2014, 18 Sco: A Solar Twin Rich in Refractory and Neutron-Capture Elements. Implications for Chemical Tagging, ApJ **791**, 14.
70. Monroe, T., Meléndez, J., Ramírez, I., Yong, D., *et al.*, 2013, High Precision Abundances of the Old Solar Twin HIP 102152: Insights on Li Depletion From the Oldest Sun, ApJ **774**, L32.

Selected Non-Refereed Publications:

71. Ford, E., **Bedell, M.**, Ciardi, D., Dodson-Robinson, S., *et al.*, 2019, *Advanced Statistical Modeling of Ground-Based RV Surveys as Critical Support for Future NASA Earth-Finding Missions*, Astro2020: Decadal Survey on Astronomy and Astrophysics, **2020**, 466
72. Meléndez, J., Bean, J., **Bedell, M.**, Ramírez, I., *et al.*, 2015, *Using Solar Twins to Explore the Planet-Star Connection With Unparallelled Precision*, The Messenger **161**, 28.